



QUESTION PAPER

Name of the Examination: Winter 2023-24 Semester Senior – CAT 1

Course Code: CSE3004

Set number: 5

Duration: 90 Mins

Course Title: Design and Analysis of Algorithms

Date of Exam: 06/02/2024 (Am)
CB29

Total Marks: 50

Q1. Prove or disprove each of the following: (10M)

(a) $13n^2 + 7n + 2 = O(n^2)$

(b) $5 \cdot 2^n + n^2 = O(2^n)$

Q2. Solve the following recurrence relations using Master's method: (10M)

(a) $T(n) = \sqrt{2} T(n/2) + \log n$

(b) $T(n) = 2 T(n/2) + n^2/\log n$

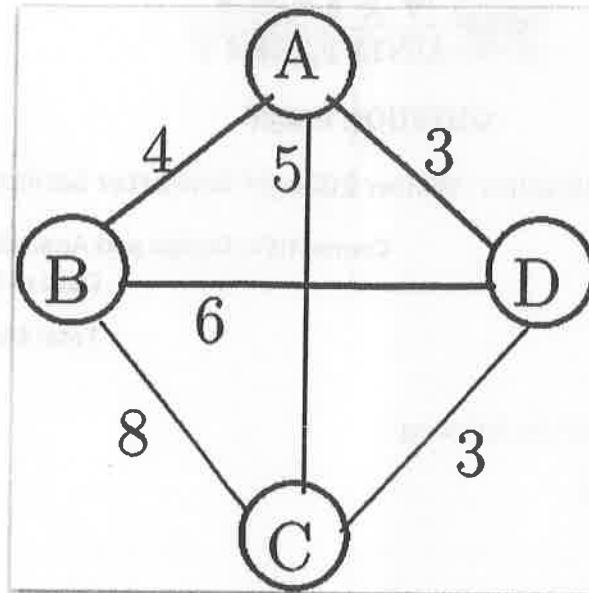
Q3. Consider the following instance of Knapsack problem: (10M)

Item	I1	I2	I3	I4	I5
Profit	9	12	11	7	10
Weight (in Kg)	7	4	7	6	6

The maximum weight of 15 Kg is allowed in the Knapsack. Find the value of maximum profit with the optimal solution of the Knapsack problem using BRUTE FORCE methodology. The assumption is that each time the item is either taken completely or is left behind; he cannot take a fractional amount of an item or take an item more than once.

Q4. Illustrate the operation of QUICK SORT on the array $A = \{30, 22, 20, 11, 17, 19, 17, 15, 21, 5\}$. Give a brief argument that the running time of PARTITION procedure on a subarray of size n is $\theta(n)$. (10M)

Q5. a) Let us say we have 4 cities namely A, B, C, and D as shown in Figure below. A salesman start from city A and comes back to it once he sold some of the items by visiting each of the intermediary cities exactly once. Find the shortest tour that the salesman should follow using BRUTE FORCE technique: (5M)



b) Suppose we are comparing implementations of insertion sort and merge sort on the same machine. For inputs of size n , insertion sort runs in $8n^2$ steps, while merge sort runs in $64n \lg n$ steps. For which values of n does insertion sort beat merge sort? (5M)

QP MAPPING

Q. No.	Module Number	COMapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,2,3,4,5,6,11	-	-	10
Q2	1	1	1,2,3,4,5,6,11	-	-	10
Q3	2	2	1,2,4,6,7,8	-	-	10
Q4	2	2	1,2,4,6,7,8	-	-	10
Q5	1,2	1,2	1,2,4,6,7,8,11	-	-	10

QUESTION PAPER

Name of the Examination: WIN 2023-24 Semester – CAT-1

Course Code: CSE3004

Course Title: Design and Analysis of Algorithms

Set No: 6

Date of Exam: 05/02/2024 (FN) (A)

Duration: 90 min

Total Marks: 50

- Q1. (a)** Consider the following recursive algorithm. Generate the recurrence relation for the given recursive algorithm and derive the time complexity in Big O notation. **(6M)**

Algorithm Rec (n)

//Input: A positive Integer n

//Output: Factorial of a number n

```
{
    If (n <= 1) return 1;
    else return (Rec (n-1) * n);
}
```

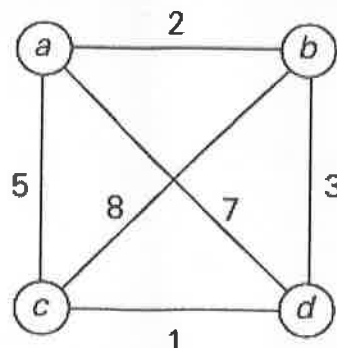
- (b)** Calculate the space complexity for the above algorithm considering the stack space and number of function calls. Find the value return by this algorithm. **(4M)**

- Q2. (a)** Solve the recurrence relation by using Masters Method.
 (i) $T(n) = 26T(n/3) + n^3$ [Given $\log_2 26 = 2.9$] (ii) $T(n) = T(n/2) + \sqrt{n}$ **(5M).**
(b) Define Big Oh and Theta Asymptotic notations and show that the following equalities are correct? **(5M)**
 (i) $\log n = \Omega(n \log n)$ (ii) $n^3 + 10n^2 = \Theta(n^3)$

- Q3. (a)** Prove that, for any two non-negative function $g(n)$ and $f(n)$, asymptotically tight bounds are obtained from asymptotic upper and lower bound i.e.

$$\Theta(g(n)) = O(g(n)) \cap \Omega(g(n)) \quad \textbf{(5M)}$$

- (b)** Solve the following travelling salesman problem using Brute force technique. **(5M)**



- Q4.** Solve the following instance of the knapsack problem with knapsack capacity 10 by the brute force algorithm: $n=4$, $(p_1, p_2, p_3, p_4) = (42, 12, 40, 25)$, $(w_1, w_2, w_3, w_4) = (7, 3, 4, 5)$. **(10M)**

- Q5.** Use Divide and Conquer technique to sort the following elements using a pivot element and perform time complexity analysis for Partition algorithm, best case and worst case.
65, 55, 70, 45, 80, 70, 35, 85, 90
(10M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	Marks
Q1	1	1	PO1, PO2, PO3, PO4, PO5, PO6, O11	5+5=10
Q2	1	1	PO1, PO2, PO4, PO6, PO7, PO8	10
Q3	1, 2	2	PO1, PO2, PO4, PO6, PO7, PO8	10
Q4	2	2	PO1, PO2, PO4, PO6, PO7, PO8	10
Q5	2	2	PO1, PO2, PO4, PO6, PO7, PO8	10